

ProCAST: A Projection Framework for Coupled Aggregation Constrained Multivariate Time Series Forecasting



Jiaqi Xue, Hongji Dong, Yucen Gao, Xiaofeng Gao*, Guihai Chen

Shanghai Key Laboratory of Scalable Computing and Systems, School of Computer Science, Shanghai Jiao Tong University, China

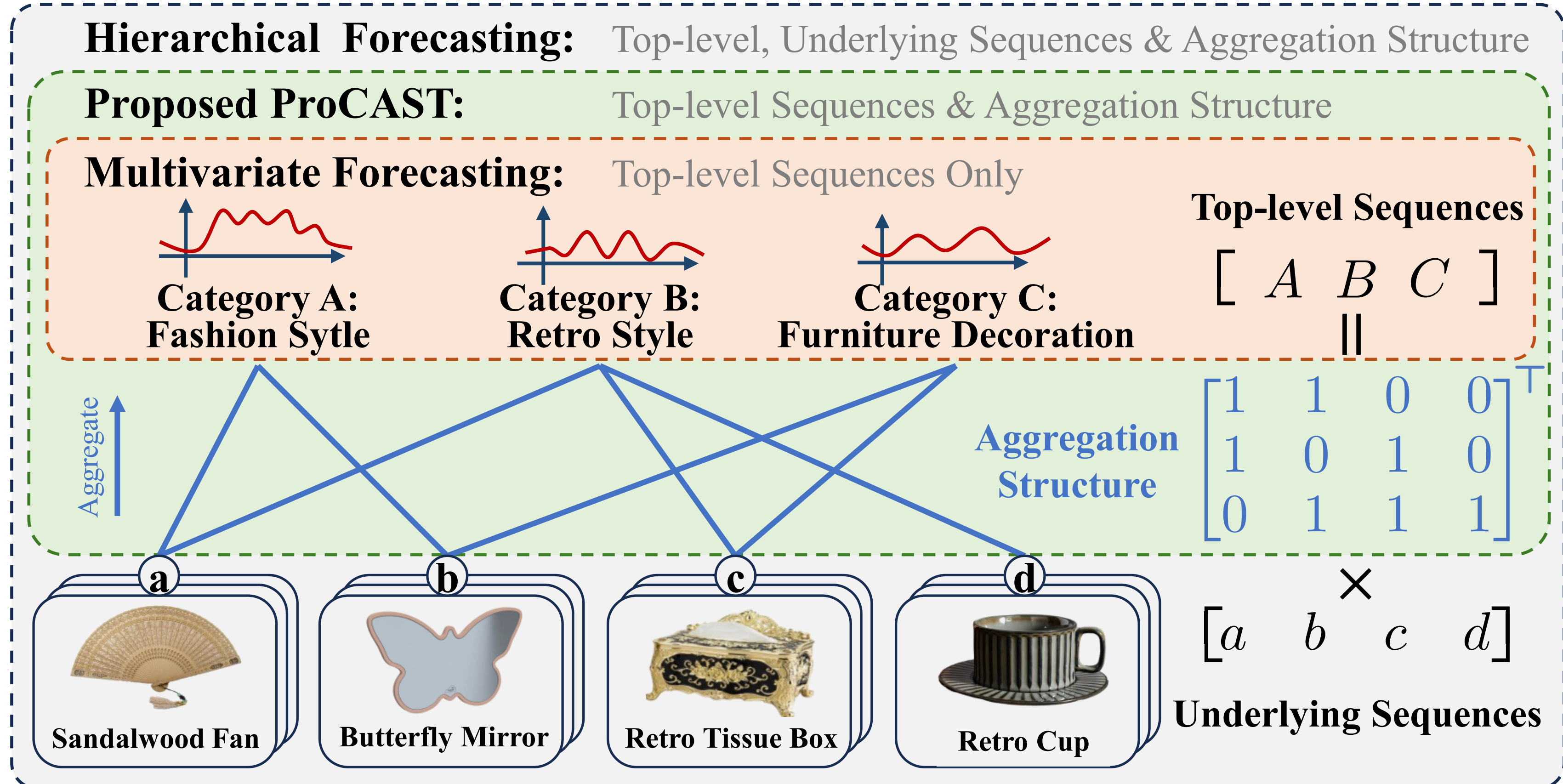
* Corresponding Author

Introduction

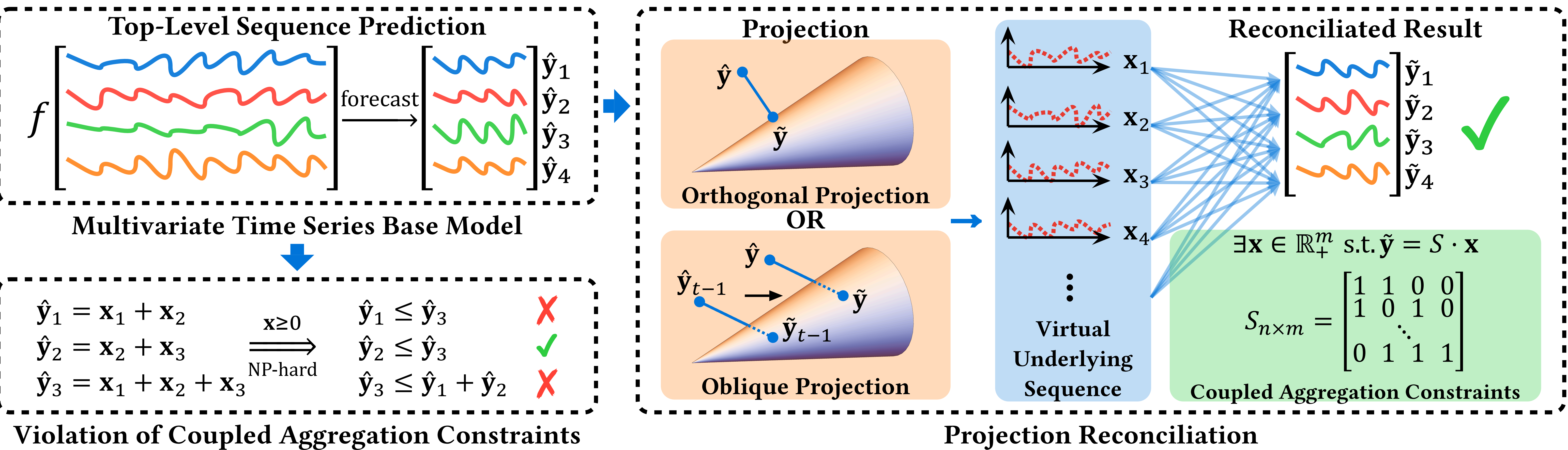
In aggregation structures, we often **focus on top-level forecasts**, as they are most relevant for sales trends, supply chain planning, and industry outlooks. When aggregation is coupled, simple non-negativity constraints on underlying sequences can **induce highly complex constraints on top-level sequences**.

Challenges ① constraints are intractable to derive; ② unconstrained forecasts rarely feasible; ③ underlying value often noisy/missing.

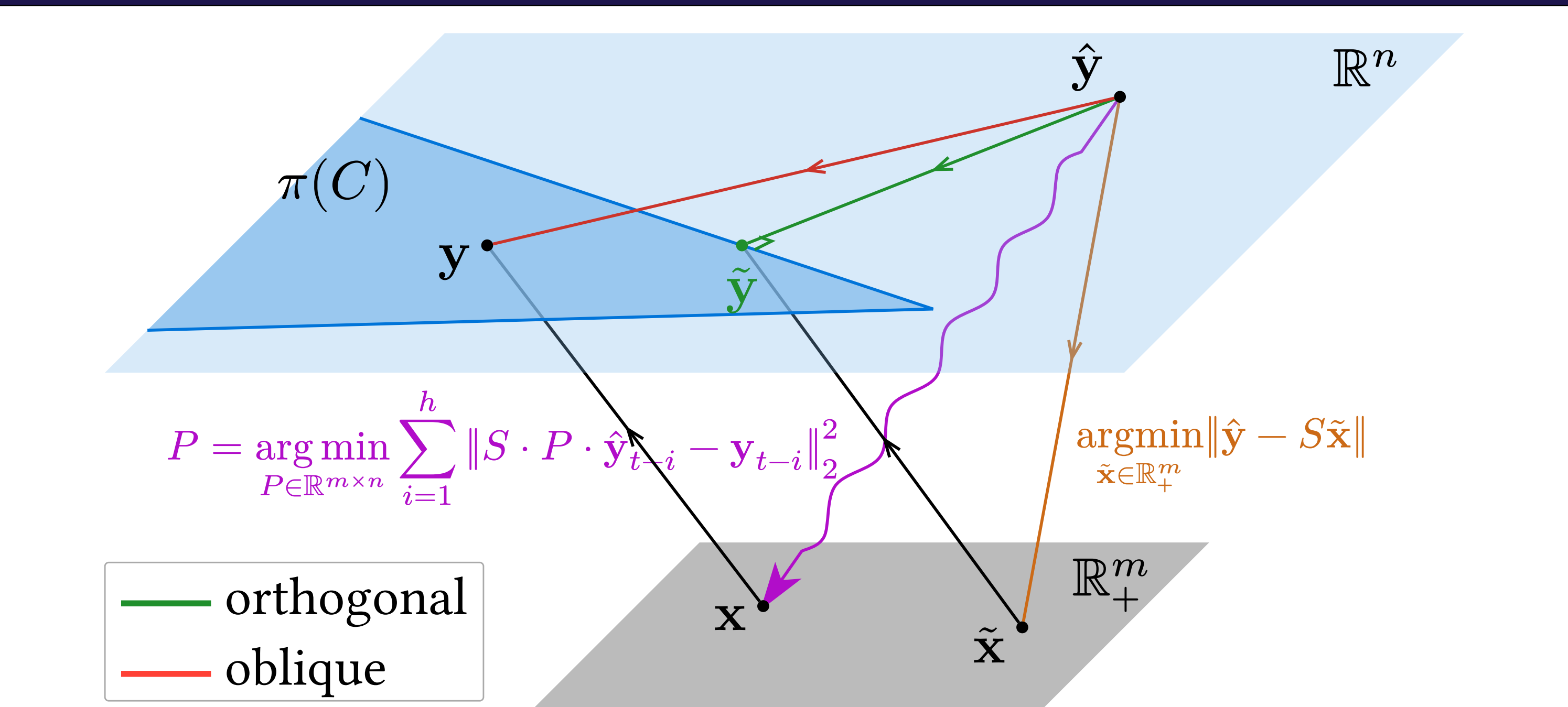
ProCAST integrates projection-based constraint enforcement into any base forecasting to deliver **accurate and feasible** top-level predictions with **theoretical guarantees**, while remaining effective without access to underlying sequences.



Overview of ProCAST, including two optional projection methods: orthogonal and oblique



Theoretical Guarantees



Orthogonal projection minimizes the Euclidean distance to the feasible set, **guaranteeing error reduction** under distance-based loss function.

$$\forall y \in \pi(C), \|y - \tilde{y}\|_2 \leq \|y - \hat{y}\|_2$$

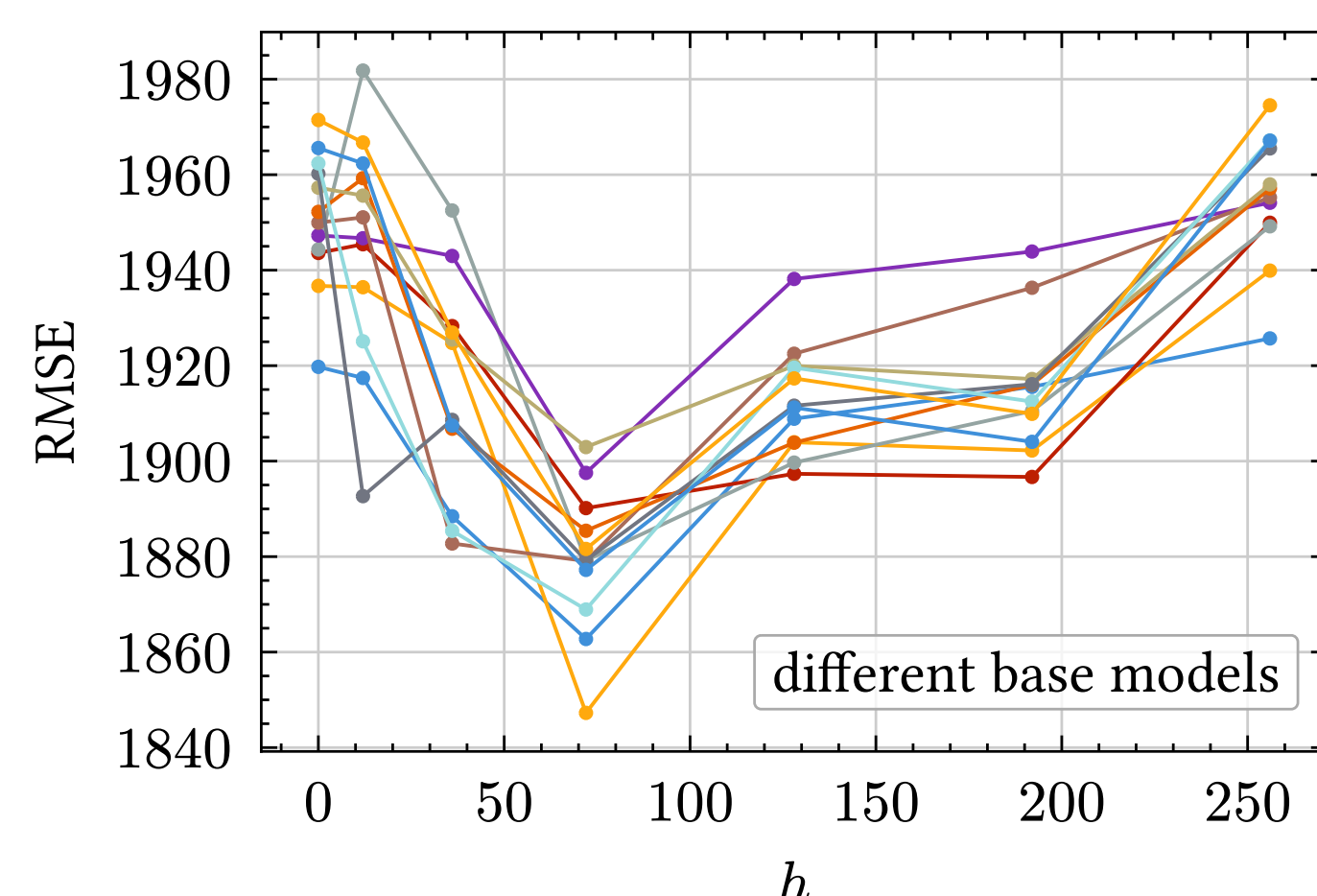
Oblique projection incorporate a data driven approach that learn a “projection matrix” from historical data, which enables the projected points to lie within the convex cone. Assuming stationarity over time $\tau \in [t - h, t]$, oblique projection **reduces the error in expectation**.

$$\mathbb{E}[\|\tilde{y}_\tau - y_\tau\|_2^2] \leq \mathbb{E}[\|\hat{y}_\tau - y_\tau\|_2^2]$$

Experiment Results

Best results are **bolded**; second-best underlined; cells shaded in gray indicate predictions that violate Coupled-Coherence.

Dataset	E-Commerce Dataset										RH Dataset					
Metric	MAE					RMSE					MAE			RMSE		
Method	Raw	BU	MinT	Ortho.	Obliq.	Raw	BU	MinT	Ortho.	Obliq.	Raw	Ortho.	Obliq.	Raw	Ortho.	Obliq.
NBEATS	1092	1149	1083	1092	1024	1983	2052	1975	1983	1877	827.3	827.3	752.7	1262	1262	1153
NBEATSx	1092	1149	1083	1092	1024	1983	2052	1975	1983	1877	827.3	827.3	752.7	1262	1262	1153
NHITS	1103	1194	1100	1103	1032	2001	2102	2002	2001	1901	867.5	867.5	807.0	1331	1331	1240
TimesNet	1050	1175	1082	1050	1038	1949	2087	1989	1949	1887	584.1	584.6	562.9	880.4	879.9	842.4
TCN	1067	1258	1086	1067	1054	1959	2155	1985	1959	1912	674.7	674.7	638.2	1013	1013	951.2
BiTCN	1073	1174	1075	1073	1069	1978	2084	1979	1978	1896	3382	3379	3046	5674	5655	5306
DeepNPTS	1058	1209	1111	1058	1039	1944	2115	2015	1944	1886	664.0	664.0	630.3	1023	1022	961.7
TFT	1065	1236	1074	1065	1048	1957	2133	1969	1957	1877	608.7	610.2	582.7	913.7	909.3	860.9
TiDE	1080	1122	1120	1080	1032	1985	2033	2030	1985	1886	1258	1258	1133	1949	1949	1762
DLinear	1456	1549	1496	1414	1319	2411	2500	2406	2330	2226	3690	3311	3283	9510	7620	7539
Informer	3053	2890	3048	3053	1406	3702	3560	3698	3702	2347	2.2e4	2.2e4	2.1e4	2.7e4	2.7e4	2.6e4
Autoformer	1050	1114	1058	1050	1052	1950	2032	1963	1950	1880	1025	1025	910.4	1554	1554	1388
FEDformer	4953	1115	1059	1053	1045	1949	2029	1958	1949	1874	595.5	595.5	571.9	901.5	901.4	854.7
PatchTST	1041	1122	1054	1041	1057	1924	2030	1945	1924	1881	589.7	589.6	561.8	864.3	864.3	817.9
TimeXer	4974	1169	1082	1074	1061	1973	2075	1981	1972	1897	799.3	799.6	741.3	1214	1213	1115
TimeMixer	1063	1149	1067	1063	1051	1966	2062	1972	1966	1887	608.3	608.8	585.9	920.9	920.5	878.1
TSMixer	1057	1140	1065	1057	1048	1959	2054	1968	1959	1877	619.0	619.0	589.0	948.6	948.6	888.6
TSMixerx	1062	1134	1067	1062	1057	1963	2047	1966	1963	1897	731.6	733.5	704.8	1123	1118	1061
iTransformer	4997	1173	1091	1097	1076	1983	2075	1981	1982	1928	612.7	612.8	595.2	917.8	916.4	883.7
RMoK	1072	1162	1076	1072	1047	1962	2069	1969	1962	1901	682.1	682.6	634.6	1037	1037	957.5
SOFTS	1106	1174	1093	1106	1091	1984	2075	1981	1984	1921	669.7	670.5	634.9	986.8	986.3	927.8
StemGNN	1059	1236	1083	1059	1055	1950	2136	1981	1950	1886	890.7	894.0	826.0	1396	1367	1258



Results: Avg. RMSE reduction **5.15%** (E-Commerce) and **7.10%** (RH); MAE reduction **4.33%** and **6.19%** respectively. ProCAST removes all coupled-coherence violations and outperforms hierarchical baselines in practice. **Stationarity is consistently validated** across multiple base forecast models.